College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

ASSIGNMENT 03 — MARKING SCHEME

Sessions Covered: 05–12 and 15 | 40 MCQs × 1 mark = 40 marks | Excluded: 13, 14

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Duration: 80 Minutes **Total Marks:** 40 **Marking:** 1 mark per correct answer. No negative marking.

Quick key (Question → Answer):

1C 2B 3B 4B 5B 6B 7C 8A 9B 10C
11B 12B 13B 14B 15B 16C 17B 18B 19B 20B
21B 22B 23B 24B 25A 26B 27A 28B 29C 30C
31B 32B 33B 34B 35B 36B 37B 38B 39B 40B

Section	Topics / Sessions	Marks
A	Methods, Paradigm & Classes (05–09) — Q1–Q14	14
B	Encapsulation & Access (10–11) — Q15–Q27	13
C	Constructors & Inheritance (12, 15) — Q28–Q40	13
Total		**40**

Section A — Methods, Paradigm & Classes (Q1–Q14)

Q	Ans	Expected answer / justification
1	C	`void` means the method returns no value.
2	B	Procedural style = step-by-step functions on shared data.
3	B	Four pillars: Encapsulation, Inheritance, Polymorphism, Abstraction.
4	B	A class is a blueprint/template for objects.
5	B	`Rectangle rect = new Rectangle();` instantiates an object.
6	B	A field is class-level data belonging to the object.
7	C	Instance methods are called on the object: `car.accelerate();`
8	A	Parameters are inputs passed into a method when it is called.
9	B	`return` ends the method and sends the value to the caller.
10	C	Return type comes before the name: `public double getArea()`.
11	B	Each object has its own separate field values.
12	B	Variables inside a method are local variables.
13	B	OOP bundles fields and methods together in a class.
14	B	Call must supply the required `double` argument.

Section B — Encapsulation & Access (Q15–Q27)

Q	Ans	Expected answer / justification
15	B	Encapsulation = hide data; expose controlled methods.
16	C	`private` = visible only inside the same class.
17	B	Public fields can be changed illegally (e.g. negative balance).
18	B	A getter returns/reads a private field safely.
19	B	A setter can validate before assigning to a private field.
20	B	JavaBeans naming: `getTitle()` for field `title`.
21	B	Only the public getter is accessible from outside; not `balance`.
22	B	Validation blocks invalid states when data is changed.
23	B	`this.balance` refers to the current object's field.
24	B	`protected` allows subclass and same-package access.
25	A	Getters/setters control how data is read and changed.
26	B	Condition fails; field is not set to −5.
27	A	Boolean getters commonly use `getX()` or `isX()`.

Section C — Constructors & Inheritance (Q28–Q40)

Q	Ans	Expected answer / justification
28	B	Constructor name = class name; no return type.
29	C	Constructor runs when the object is created with `new`.
30	C	Child class uses `extends` Parent.
31	B	Inheritance models an **is-a** relationship.
32	B	`Car extends Vehicle` → inherits non-private members.
33	B	`super(...)` calls the parent constructor.
34	B	Java supplies a default no-arg constructor if none is written.
35	B	Valid constructor: same name as class, no return type (not even `void`).
36	B	Access private parent data via getters/setters or protected members.
37	B	Java allows single class inheritance only.
38	B	A `Car` is-a `Vehicle`, so `Vehicle v = new Car();` is valid.
39	B	`super(...)` must be the first statement in the constructor.
40	B	`Dog`'s `speak()` runs → prints Woof.

Accept only the single best option shown. If a student marks two options for one question, award 0 for that question.

******* END OF MARKING SCHEME *******